

- 1 A ball is placed at the top of a slope and a block is fixed rigidly to the lower end of the slope as shown in Fig. 1.1.

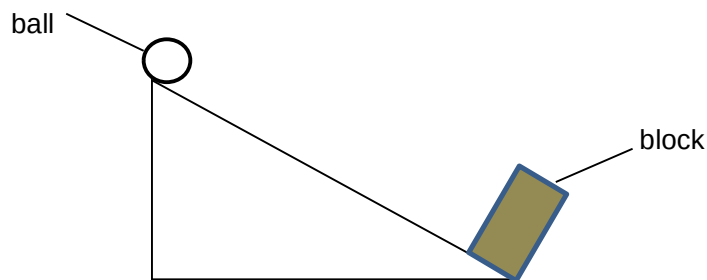


Fig. 1.1

The ball is released at time $t = 0$ from the top of the slope and v , the velocity of the ball down the slope, is found to vary with t as shown in Fig. 1.2.

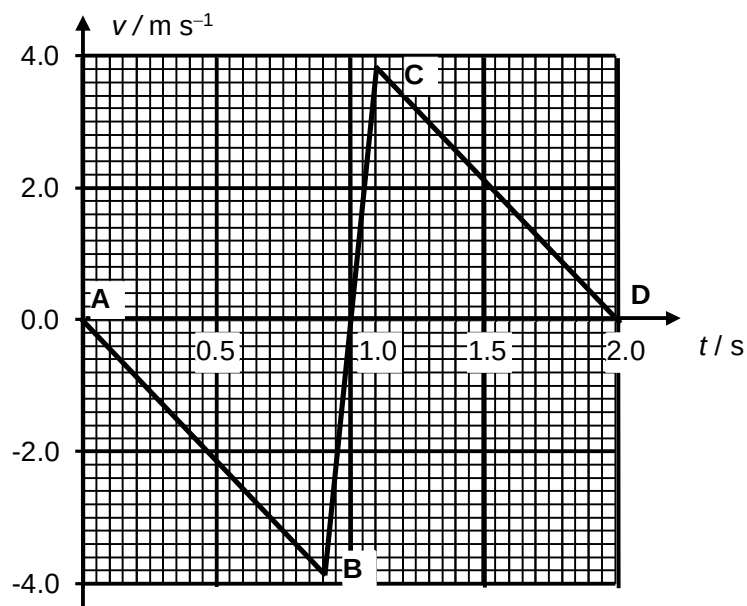


Fig. 1.2

- (a) Describe qualitatively the motion of the ball during the periods AB, BC and CD.

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..... [3]

(b) Calculate

(i) the acceleration of the ball during impact with the block, and

acceleration = m s^{-2} [2]

(ii) the length of the incline just before contact with the block.

length = m [1]

2 A McLaren Formula 1 car is fitted with a rocket engine that consumes fuel at a rate of